1. Objective

The **PixelDwell AI Photo Editing Platform** aims to deliver a scalable, AI-powered image processing system tailored for real estate workflows.

The platform will transform raw property images into high-quality, market-ready visuals by combining:

- Deterministic computer vision techniques (HDR, alignment, corrections)
- Python-based processing pipelines
- AI-powered enhancement services for advanced transformations

The solution is designed to ensure **rapid deployment, high output quality, and long-term scalability**.

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2. Solution Strategy

The system follows a **hybrid modular architecture** built on three key principles:

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1. Algorithmic Core (Python-Based)

- Handles:
  - HDR generation
  - image alignment
  - base corrections
- Built using:
  - OpenCV
  - scikit-image

---

2. Library-First Approach

- Prefer existing Python packages wherever feasible
- Reduces:
  - development time

    - complexity
- Ensures:
    - reliability
    - performance

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3. AI Enhancement Layer (Pluggable)

- Used for complex, non-deterministic features
- Powered initially by:
    - **Google Nano Banana API (representative external AI service)**
- Designed to:
    - evolve into in-house models over time

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3. End-to-End Workflow

Workflow Overview

[Image Upload]
↓
[EXIF Extraction + Fuzzy Matching]
↓
[Scene Clustering]
↓
[Image Alignment]
↓
[HDR Fusion Engine]
↓
[Base Enhancements]
↓
[AI Enhancement Layer]
↓
[Final Output]

🔍 Detailed Processing Pipeline

Input Images

```
  │
  ├── Metadata Extraction (EXIF)
  ├── Perceptual Hashing (pHash)
  │
  ↓
Scene Grouping Engine
  │
  ↓
Alignment Engine (OpenCV)
  │
  ↓
HDR Engine (Mertens Fusion)
  │
  ↓
Base Enhancement Layer
  │
  ├── White Balancing
  ├── Color Correction
  ├── Sharpening
  ├── Lens Correction
  │
  ↓
Enhancement Orchestrator
  │
  ├── Local Processing (OpenCV)
  └── API Calls (Nano Banana)
  │
  ↓
Final Image Output
```

---

4. 🧩 Feature Implementation Strategy

This section defines how each feature category will be implemented using a **layered execution model**.

🟢 A. Algorithm-Based (Local Python / OpenCV)

These features are implemented using deterministic computer vision techniques:

- Auto HDR & Bracketing (Exposure Fusion – Mertens)
- HDR Image Enhancement
- White Balancing
- Lighting & Colour Correction
- Image Sharpening
- Lens Correction
- Perspective Correction
- Auto Vertical Correction
- Vertical & Horizontal Alignment
- Window Pull & Sky View Enhancement

Benefits:

- Fast execution
- No external dependency
- High consistency

🟡 B. Library + Rule-Based Hybrid

These features use a mix of OpenCV and rule-based logic:

- Custom Style Profiles (filters, LUTs)
- Lawn Enhancement / Greening (color segmentation)
- Fire Replacement (template blending)
- Basic decluttering (mask-based)

Benefits:

- Controlled output
- Reduced API usage

🔴 C. API-Based Enhancements (Phase 1)

For advanced or generative transformations, external AI APIs will be used:

- Sky Replacement
- Object Removal (AI-based)
- Decluttering (AI-based)
- Virtual Staging (AI/CGI)
- Day-to-Dusk Conversion

- Virtual Renovation Sketch Overlay

Implementation:

- Integrated via **Google Nano Banana API (example external service)**

Benefits:

- Faster deployment
- High-quality results
- No need for model training

🔁 D. Future In-House AI Models

Over time, API-based features can be replaced with:

- Segmentation models (U-Net, DeepLab)
- Inpainting models (object removal)
- Diffusion models (staging, renovation)

Benefits:

- Cost optimization
- Full control
- Customization

---

5. 🧠 Enhancement Decision Logic

The system dynamically determines how to process each feature:

```
For each enhancement request:
  │
  ├── Is feature supported by OpenCV / Python?
  │       ├── YES → Process Locally
  │       └── NO  → Call AI API
  │
  ↓
Return Enhanced Output
```

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6. Complete Feature Coverage

Core HDR & Processing

- AI Photo Editing (Auto HDR, Bracketing)
- HDR Image Enhancement

Color & Lighting

- White Balancing
- Lighting & Colour Correction
- Image Sharpening
- Custom Style Profiles

Geometry & Structure

- Perspective Correction
- Auto Vertical Correction
- Vertical & Horizontal Alignment
- Lens Correction

Exposure Enhancements

- Window Pull & Sky View Enhancement

Environment Enhancements

- Sky Replacement
- Lawn Replacement / Greening
- Fire Replacement

Object-Level Editing

- Decluttering (AI/manual)
- Item/Object Removal (AI-based)

Scene Transformation

- Day to Dusk Conversion
- Virtual Staging (AI/CGI)
- Virtual Renovation Sketch Overlay

Floor Plan & Spatial Features

- 2D Black & White Floor Plan
- 2D Colored Floor Plan
- 3D Floor Plan
- Floor Plan Conversion (2D ↔ 3D)
- Site Map / Plot Map Overlays
- Matterport Integration (optional)

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7. ⚙️ Technology Stack (Backend-Focused)

🐍 Core Language:

- Python

📦 Libraries:

Computer Vision:

- OpenCV
- scikit-image

Image Processing:

- Pillow
- imagehash

Metadata:

- exifread

🤖 AI Integration:

- Google Nano Banana API (external AI service)

⚙️ Backend (Optional):

- FastAPI (API layer)
- Celery (async processing)

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8. 🧩 Architecture Overview

```
Core Processing Engine (Python)
  │
  ├── Image Grouping Module
  ├── Alignment Engine
  ├── HDR Engine
  ├── Base Enhancement Modules
  │
  ↓
Enhancement Layer
  │
  ├── Local Processing (OpenCV आधारित)
  │
  └── API Adapter Layer
```

└── Google Nano Banana API

---

9. 🚀 Key Advantages

⚡ Faster Time-to-Market

- Prebuilt libraries + APIs reduce development effort

🧠 Intelligent Processing

- Combines deterministic and AI approaches

🔄 Scalable Architecture

- Easy transition from API → in-house models

💰 Cost Optimization

- APIs used selectively

🎯 High Output Quality

- HDR precision + AI enhancements

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11. 📌 Conclusion

The **PixelDwell AI Photo Editing Platform** delivers a robust and scalable solution by:

- Implementing **core HDR and correction pipelines using Python-based algorithms**

- Leveraging **existing computer vision libraries wherever feasible**
- Integrating **AI services (e.g., Google Nano Banana API)** for advanced enhancements
- Ensuring a **modular and future-ready architecture**

This approach enables **rapid deployment, high-quality outputs, and long-term scalability**, making it well-suited for modern real estate imaging workflows.